

JKBOSE

2026 CLASS 11TH

NEWTON'S LAWS OF MOTION

LECTURE-1

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1 Revision ✓
- 2 Inertia ✓
- 3 1st Law of Motion ✓
- 4 Momentum ✓
- 5 2nd Law of Motion ✓
- 6 Impulse ✓

Mehnat Karo

Recap from 1st Lecture

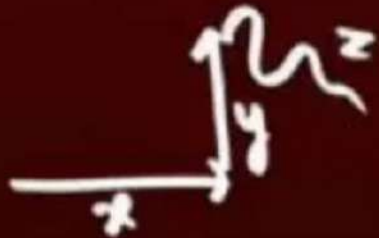
- Motion in 1D



- Motion in 2D



- Motion in 3D



- Displacement in 2D

$$\vec{r} = \vec{x} + \vec{y}$$

- Velocity in 2D

$$\vec{v} = \vec{v}_x + \vec{v}_y$$

- Acceleration in 2D

$$\vec{a} = \vec{a}_x + \vec{a}_y$$

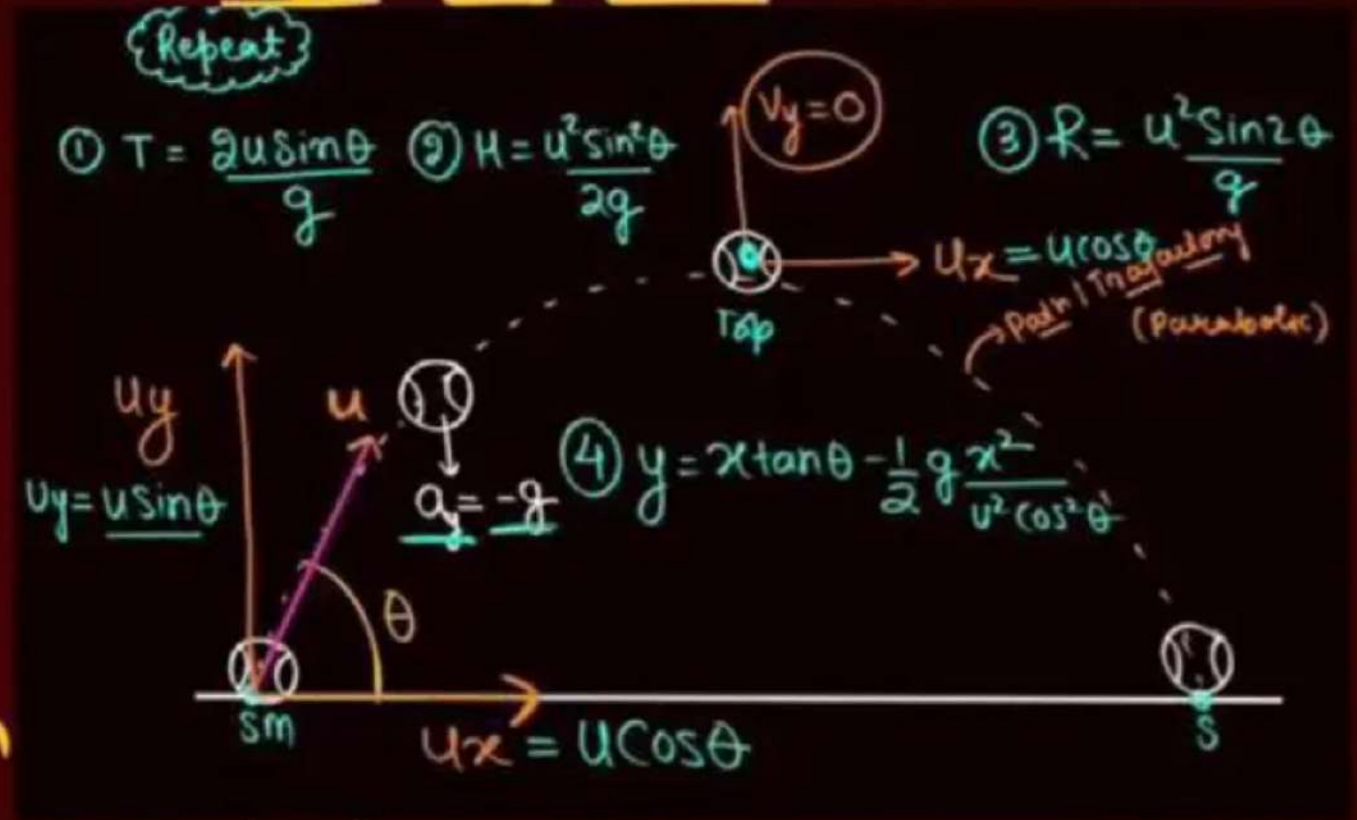
- Equation of motion

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as$$

$$v_x = u_x + a_x t \quad s_x = u_x t + \frac{1}{2}a_x t^2 \quad v_x^2 = u_x^2 + 2a_x s_x$$

$$v_y = u_y + a_y t \quad s_y = u_y t + \frac{1}{2}a_y t^2 \quad v_y^2 = u_y^2 + 2a_y s_y$$

Projectile Motion



Projectile Motion

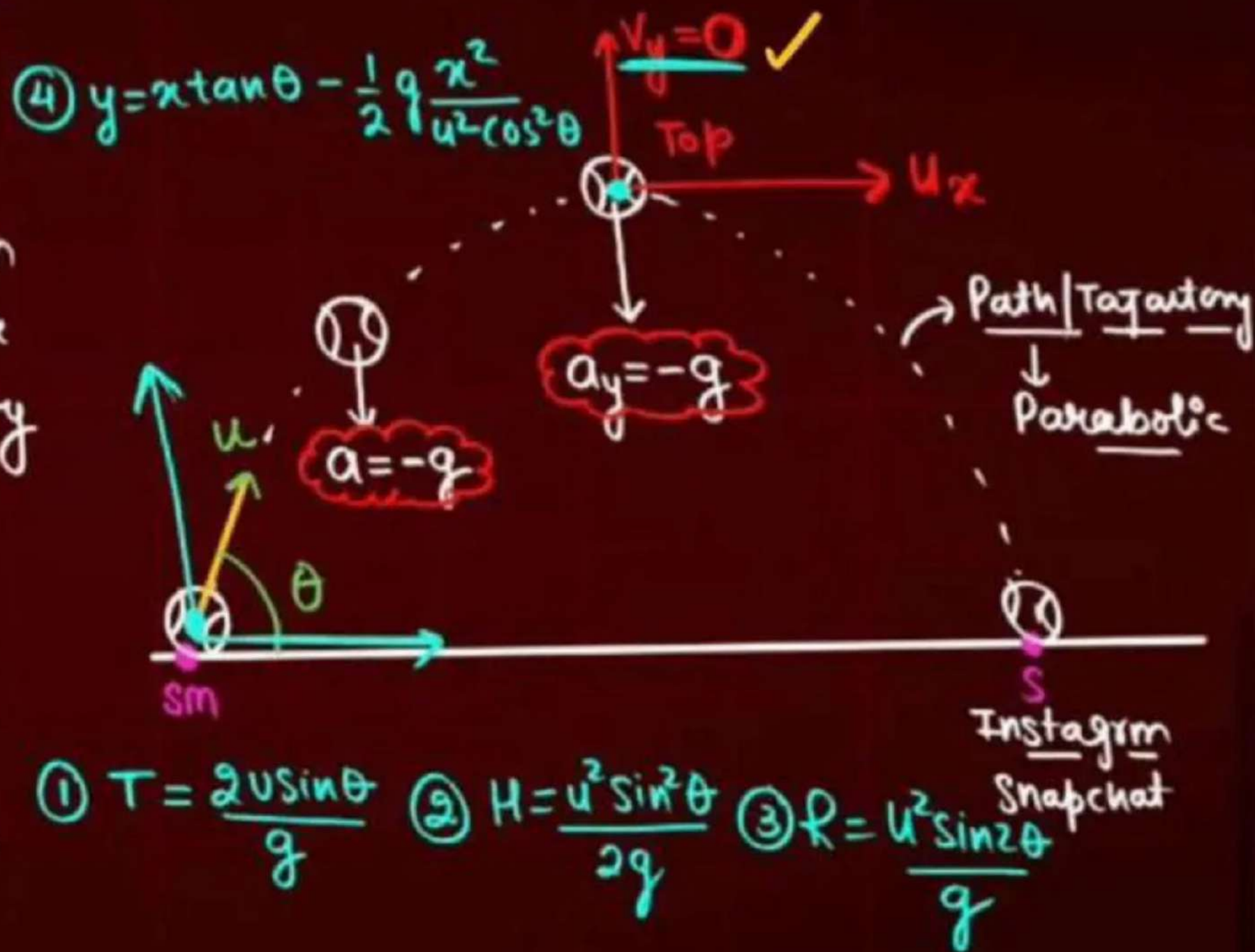
(5-mark) 2026, 2025, 2024, 2023, 2022



- Example of 2D
- To explain the 2D motion
- when an object is thrown with some Angle under the influence of gravity

$a_y = -g = -9.8 = -10 \text{ m/s}^2$
 # $a_x = 0$

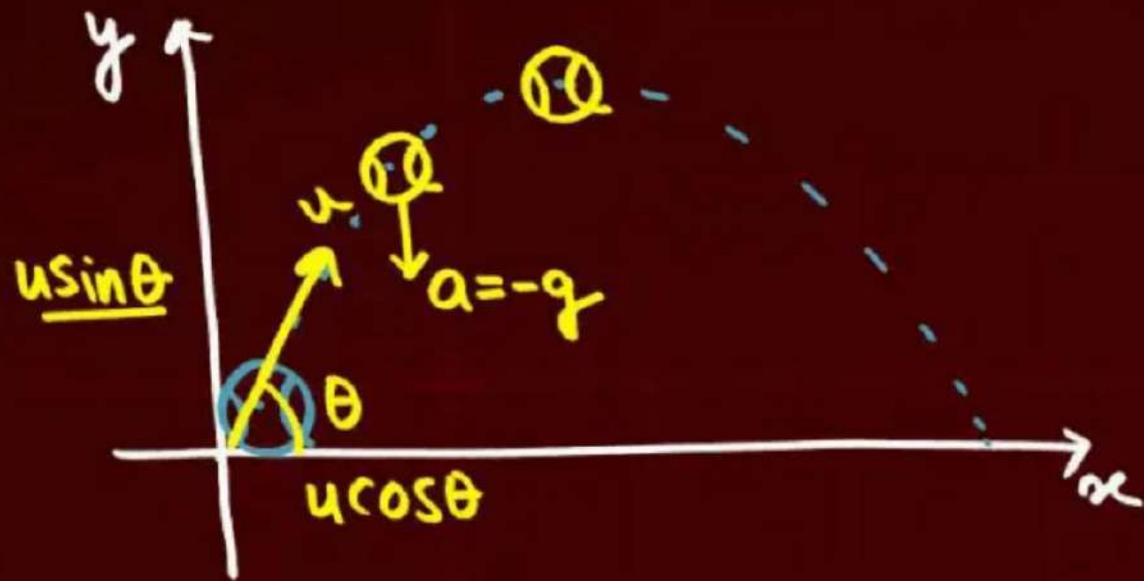
$v_y = 0$
 Path | Trajectory $\rightarrow$ Parabolic



Recap from 2nd Lecture



Projectile Motion



① Time period.

The time for which ball remain in Air $T = \frac{2u \sin \theta}{g}$

② Maximum Height.

The maximum height attain by Projectile

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

③ Range

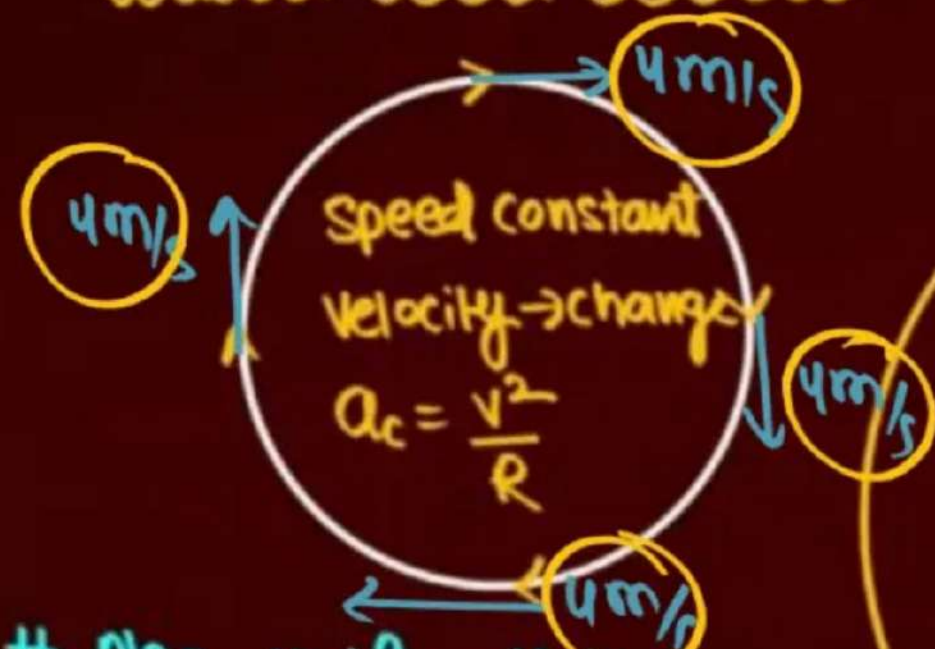
The maximum Horizontal displacement of object $R = \frac{u^2 \sin 2\theta}{g}$

④ Equation of Trajectory $\rightarrow$ Relationship b/w y and x

$$y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

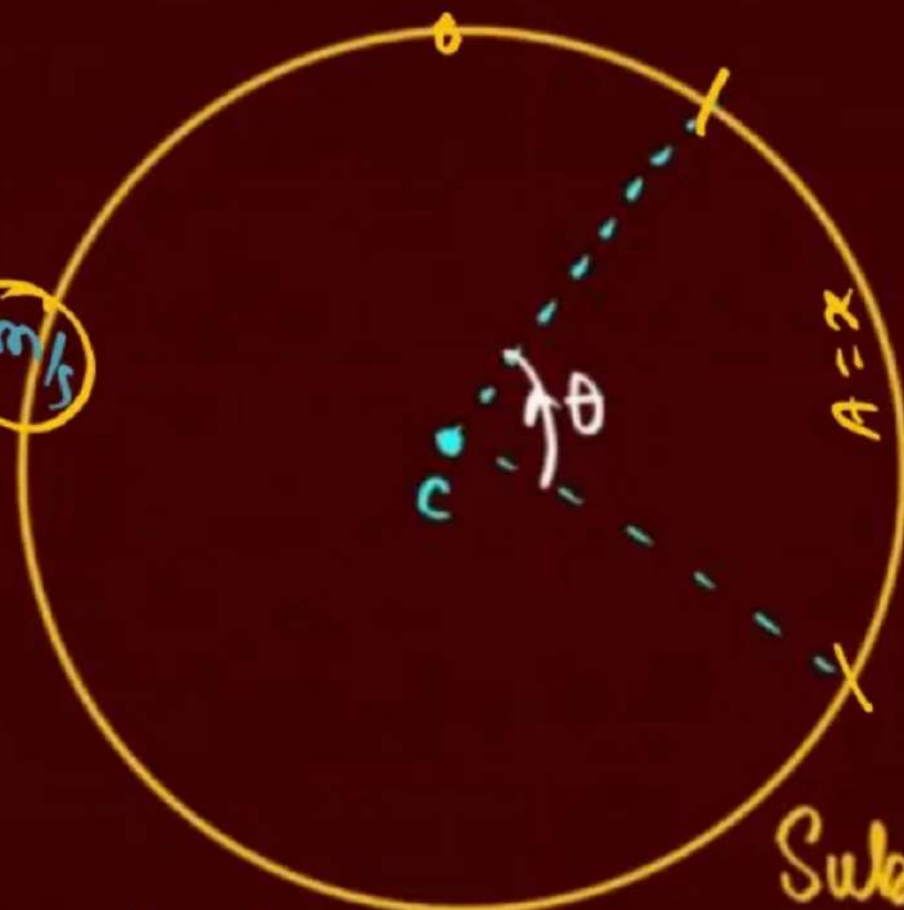
Recap from 3rd Lecture

Uniform Circular Motion



Non-uniform Motion

Speed = $\frac{2\pi r}{T} = 2\pi r \gamma$



$A + B$
 $\vec{A} - \vec{B}$

$R \quad R$

① Angular displacement

$$\theta = \frac{A}{r}$$

$$r \theta = A$$

$$\theta = \frac{A}{r}$$

② Angular velocity (ω)

$$r \omega = v$$

③ Angular acceleration (α)

$$r \alpha = a$$

Subtraction of two vector

$$R = \sqrt{A^2 + B^2 - 2AB \cos \theta} \quad \tan \alpha = \frac{B \sin \theta}{A - B \cos \theta}$$

Newton laws of Motion

- ① Inertia (Newton 1st law)
- ② Newton 2nd law (momentum) (Impulse)
- ③ Newton 3rd law (Application)
- ④ law of conservation of momentum.
- ⑤ Types of force.
- ⑥ Friction
- ⑦ centripetal force.
- ⑧ Turning on Banked Road.

Laws of Motion

Intuitive concept of force, Inertia, Newton's first law of motion; momentum and Newton's second law of motion; impulse; Newton's third law of motion. Law of conservation of linear momentum and its applications. Equilibrium of concurrent forces, Static and kinetic friction, laws of friction, rolling friction, lubrication. Dynamics of uniform circular motion: Centripetal force, examples of circular motion (vehicle on a level circular road, vehicle on a banked road).

Inertia

→ 9th (Sustii)

The property of body by virtue of which It does not change.

- Its state of Rest ✓
- Its State of Motion ✓
- Its State of Direction ✓

Until and unless any external force is applied

↑↑↑ Inertia $\propto$ mass ↑↑↑



Types Of Inertia

01

Inertia of Rest



02

Inertia Of Motion



03

Inertia Of Direction



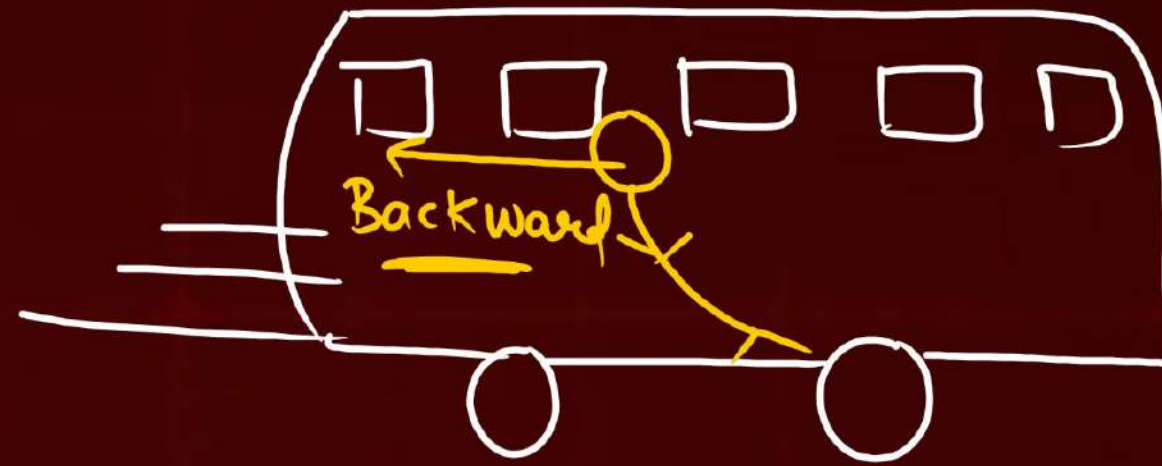
① Inertia Of Rest



If the body is in rest It will remain in rest until and unless any external force is applied ✓



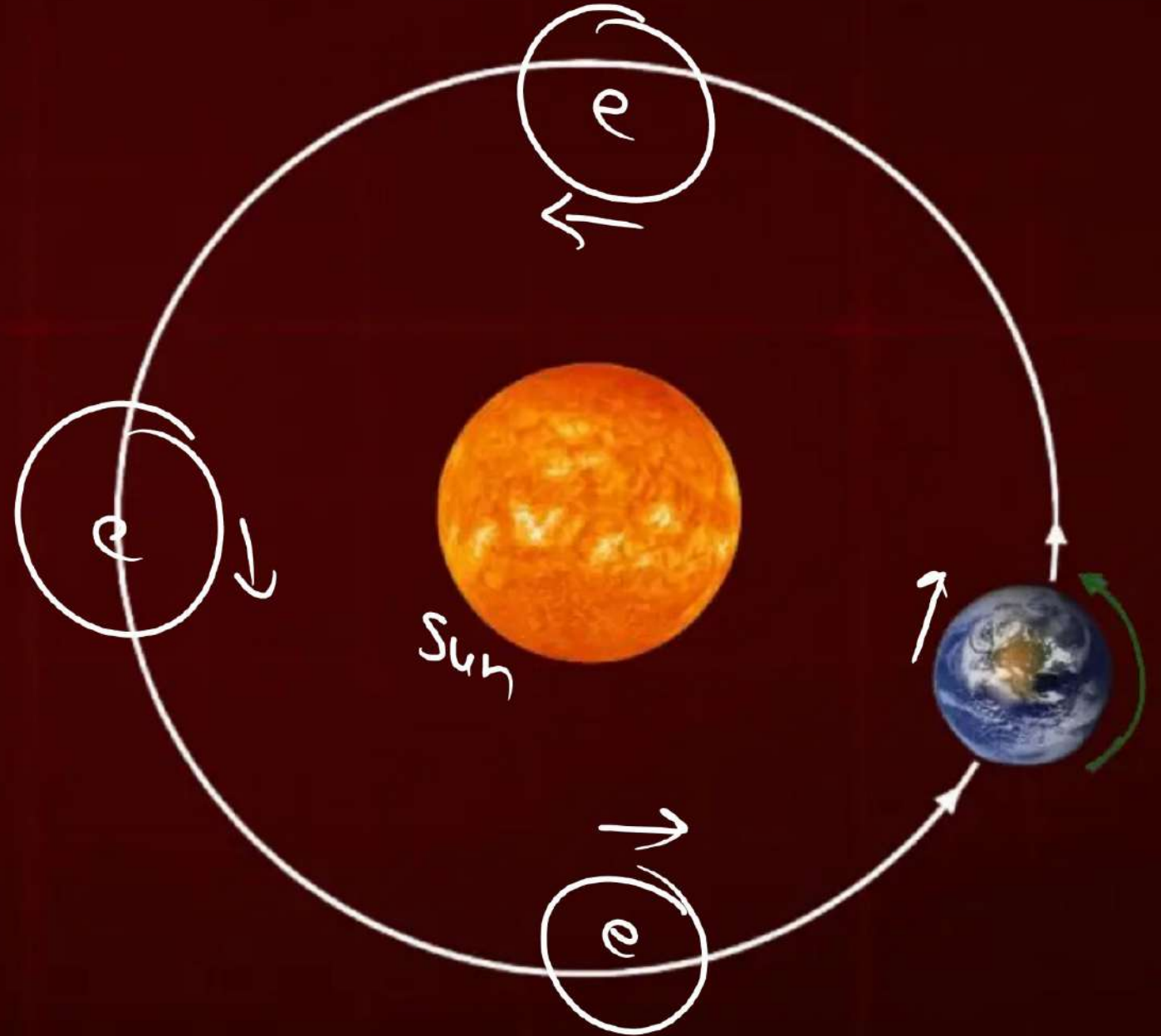
(Example of Inertia of Rest)



we feel the force in backward direction
When Body Start Suddenly
Due to Inertia of Rest ✓

② Inertia of Motion

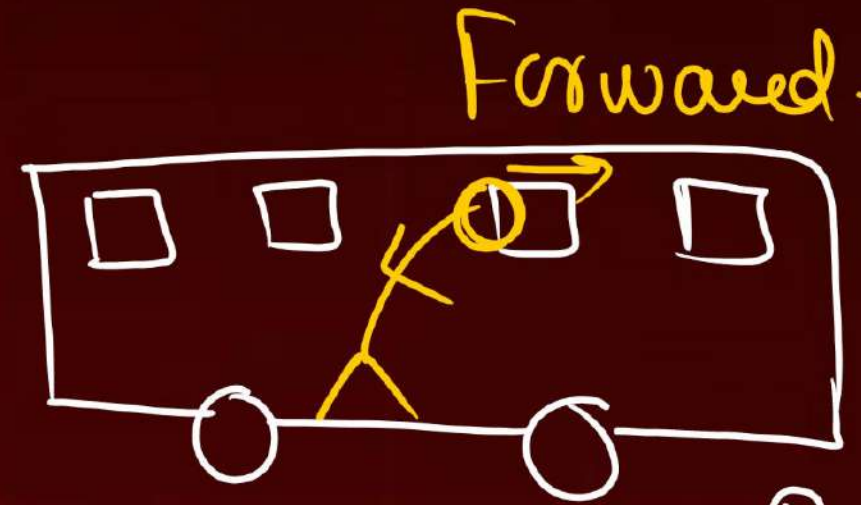
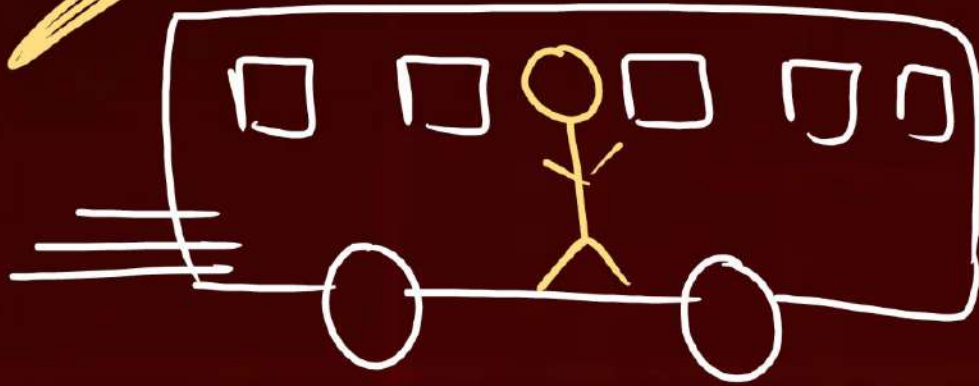
If the body is in motion it will remain in motion until and unless any external force is applied



Example of Inertia of Motion.



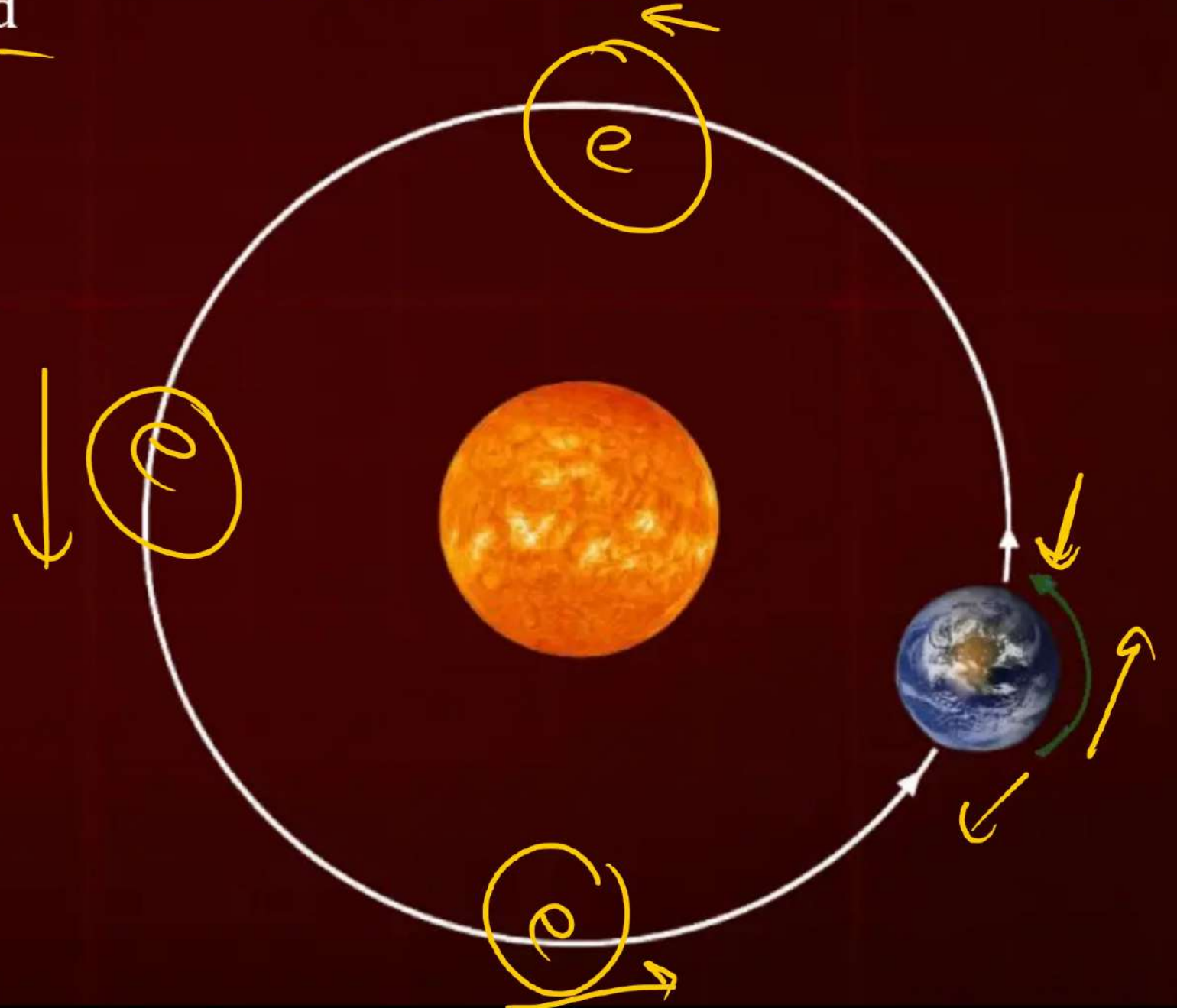
Motion



If the Body Stop Suddenly
we feel the force in forward direction
Due to inertia of motion

Inertia of Direction

If the body is in particular Direction it will remain in that particular direction until and unless any external force is applied



1st Law of Motion

Was given by Newton chacha

* Inertia is the 1st law of motion

* law of Susti

* law of inertia/mass

Q. which of the following have more inertia? (2015) Board ✓

Sol

- (i) $R_s 1$ (ii) $R_s 2$ (iii) $R_s 5$

~~(iv)~~



LPG cylinder

↑↑↑ Inertia & mass ↑↑

Momentum (P) = mv

* It give the Quantity of Motion in a Body

* Rokna = mv

* Motion = mv

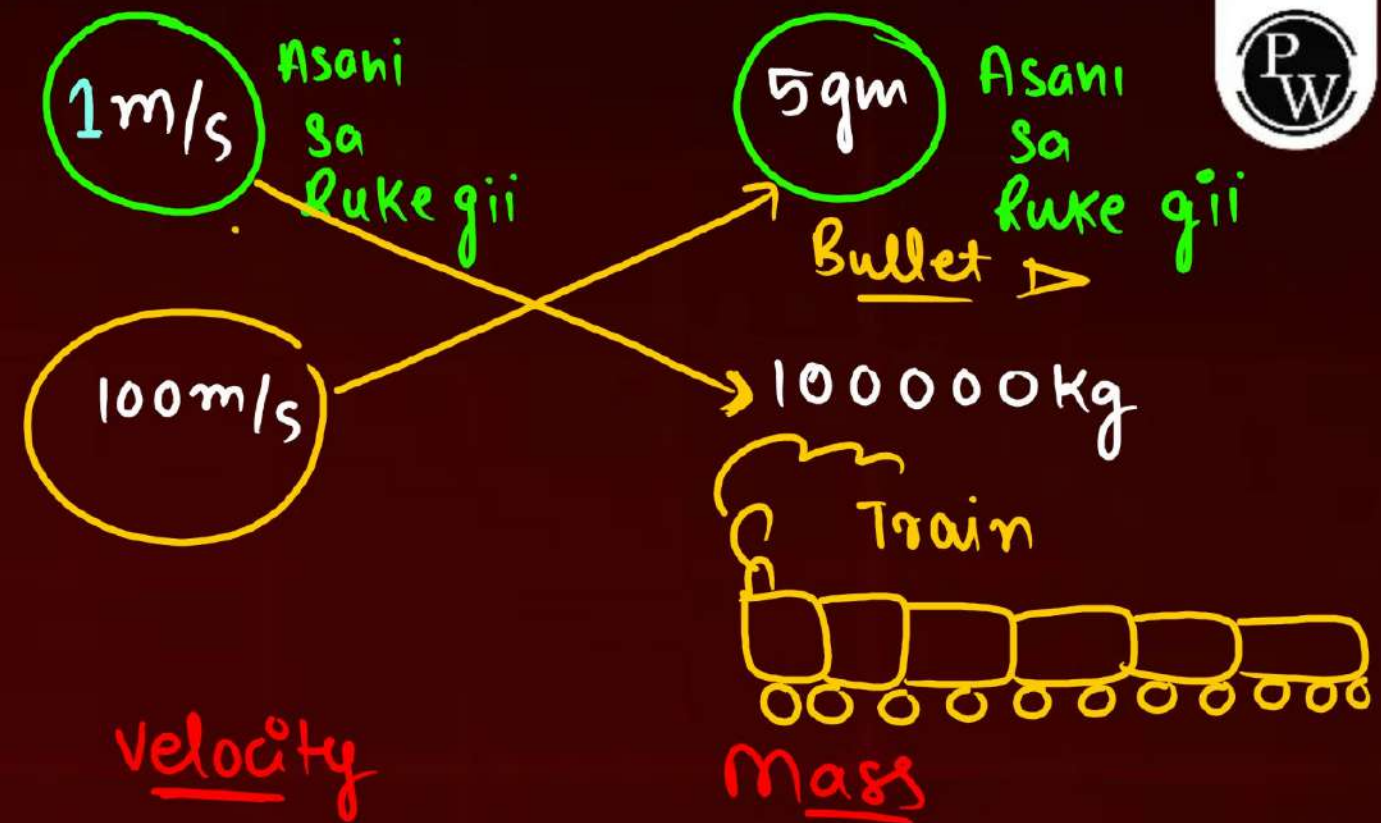
* Momentum = mv

$$\vec{P} = m\vec{v}$$

* Vector

* unit $P = mv$
 $= \text{kg m/s} \quad (\underline{\text{m(c)}})$
2024

sawal
= 0



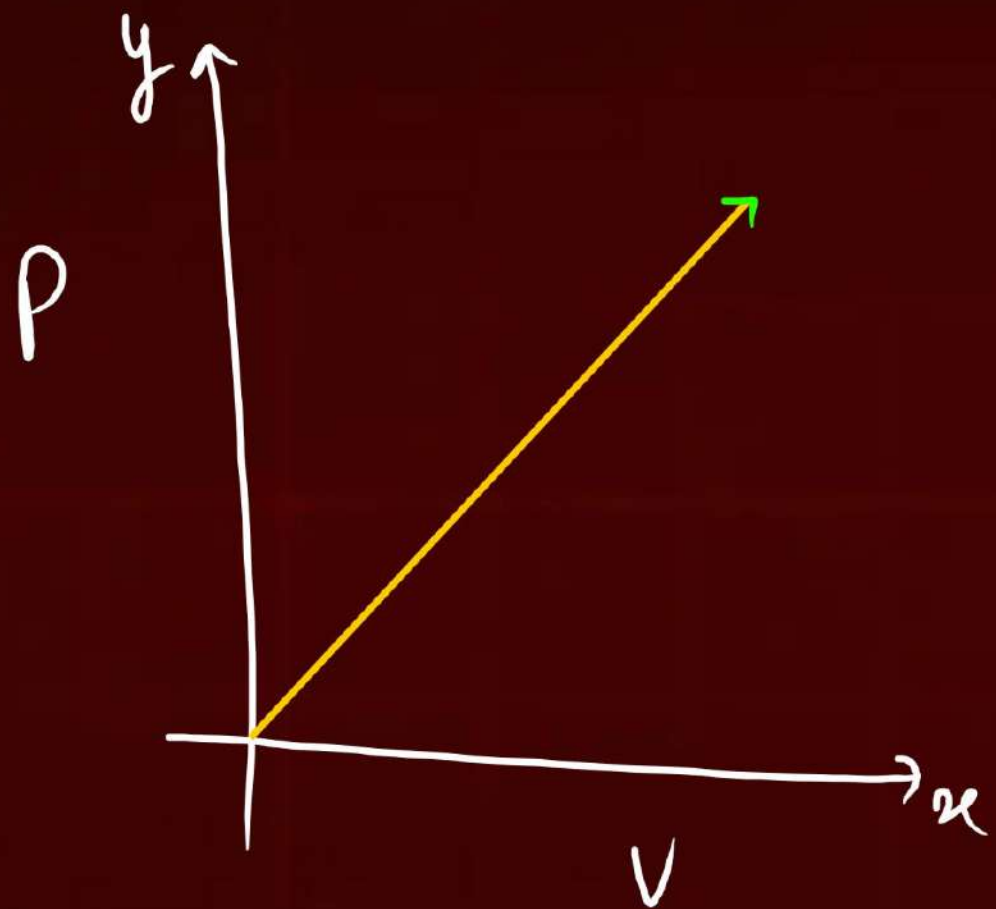
$P_i = mu$ (initial momentum)

$P_f = mv$ (final momentum)

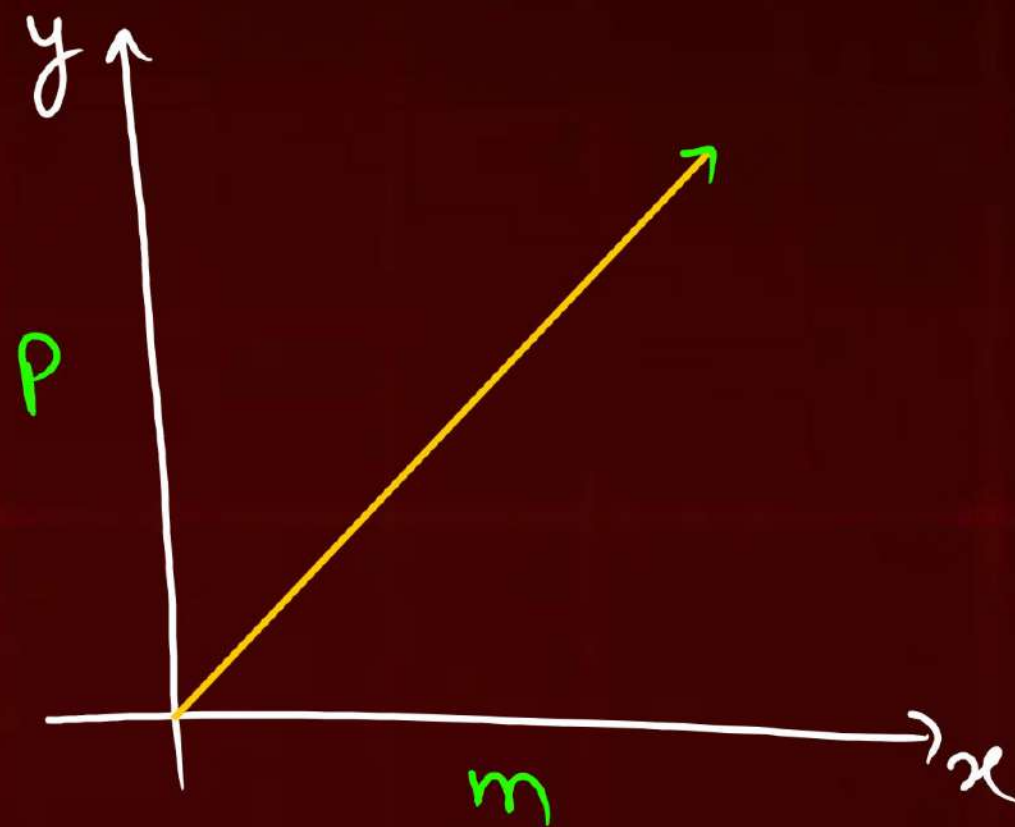
$\Delta P = P_f - P_i$

$\downarrow$
change in momentum

$\uparrow P = m V \uparrow$ (2016)



$\uparrow P = \uparrow m V$ (2016) ✓



Momentum (P)

(2017) JK Boss



Q. A Body of mass 2kg moves with velocity 10m/s find out the momentum?

$$\begin{aligned} P &= mv \\ &= 2 \times 10 \\ &= 20 \text{ kg m/s} \end{aligned}$$

(a) 10 kg m/s

☒ (b) 20 kg m/s

(c) 30 kg m/s

(d) 40 kg m/s



2nd Law Of Motion

9th → 9th
11th

$$F = m \left(\frac{v - u}{t} \right)^a$$

1st equation of Motion



9th

* It Explain about the Force

* definition of force.

* Formula of Force

$$\vec{F} = m \vec{a}$$

$$v = u + at$$

$$v - u = at$$

$$\frac{v - u}{t} = a$$

Vector

Force is Rate of change of momentum
 ΔP
 $\div t$

$$F = \frac{\Delta P}{t}$$

$$f = \frac{P_f - P_i}{t}$$

$$f = \frac{mv - mu}{t}$$

S.I unit

$$\begin{aligned} F &= ma \\ &= \text{kg m/s}^2 \\ &= \underline{\text{Newton}} \end{aligned}$$

c.g.s unit

$$\begin{aligned} F &= \underline{ma} \\ &= \text{gm cm/s}^2 \\ &= \text{dynes} \end{aligned}$$

$$1\text{N} = 10^5 \text{ dynes}$$

⑪th

"Force is Rate of change of momentum"

$$F = \frac{\Delta P}{\Delta t}$$

delta
Bada change

$$F = \frac{dP}{dt}$$

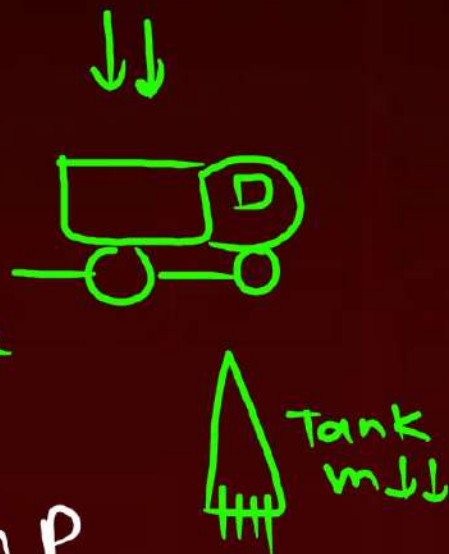
chota change in P
chota change in t

$$P = mv$$

$$F = \frac{d(mv)}{dt} = v \frac{dv}{dt} + v \frac{dm}{dt}$$

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

Real
form of
Newton 2nd law of motion.



Special case

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

$$F = m \frac{dv}{dt} + 0$$

* *

$$F = ma$$

$$\begin{aligned} \underline{dm} &= \underline{m_f} - \underline{m_i} \\ &= 60 - 60 \\ &= 0 \end{aligned}$$



Pata nahi yeh padhai ka idea kiska tha,
aaram se gufaon main rehte, patthar se
aag jalate, jhingalala huhu karte or vahi
mar jate.



What is Force

Push and Pull is called Force. ^x

Force is Rate of change of Momentum ✓ *Newton 2nd law.*

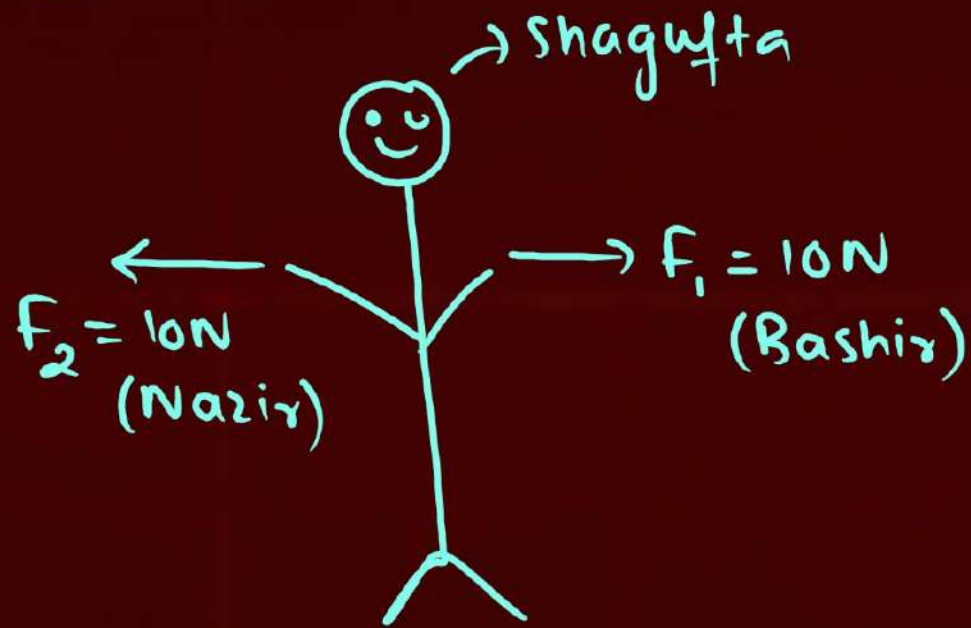
$$F = \frac{\Delta p}{t} = \frac{\Delta p}{\Delta t}$$

$$F = \frac{dp}{dt}.$$

What is equilibrium

2023

When two equal and opposite force acting on a body the $F_{\text{net}} = 0$ and body is said to be in equilibrium.



$$\begin{aligned}\vec{F}_{\text{net}} &= \vec{F}_1 + \vec{F}_2 \\ &= 10 + -10 \\ &= 10 - 10 \\ &= 0\text{N}\end{aligned}$$

Impulse

The force acting for due to short period of time.

$$\begin{array}{c} \uparrow \uparrow \uparrow F = \frac{dp}{dt} \downarrow \downarrow \downarrow \downarrow \downarrow \\ \text{Impulse} \end{array} \quad \text{or}$$

It is product of F and dt

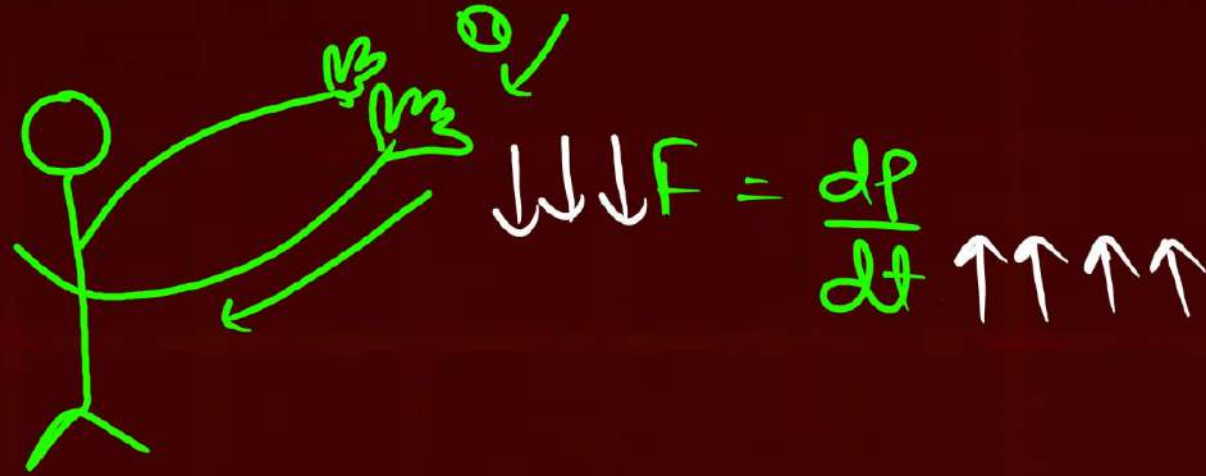
$$\uparrow F = \frac{dp}{dt} \downarrow$$

$$\text{Impulse} = \underbrace{F \times dt} = \underline{dp}$$



Application of Impulse

1. Lowering of hand while catching the Ball



Application of Impulse

2. Jumping from high Building

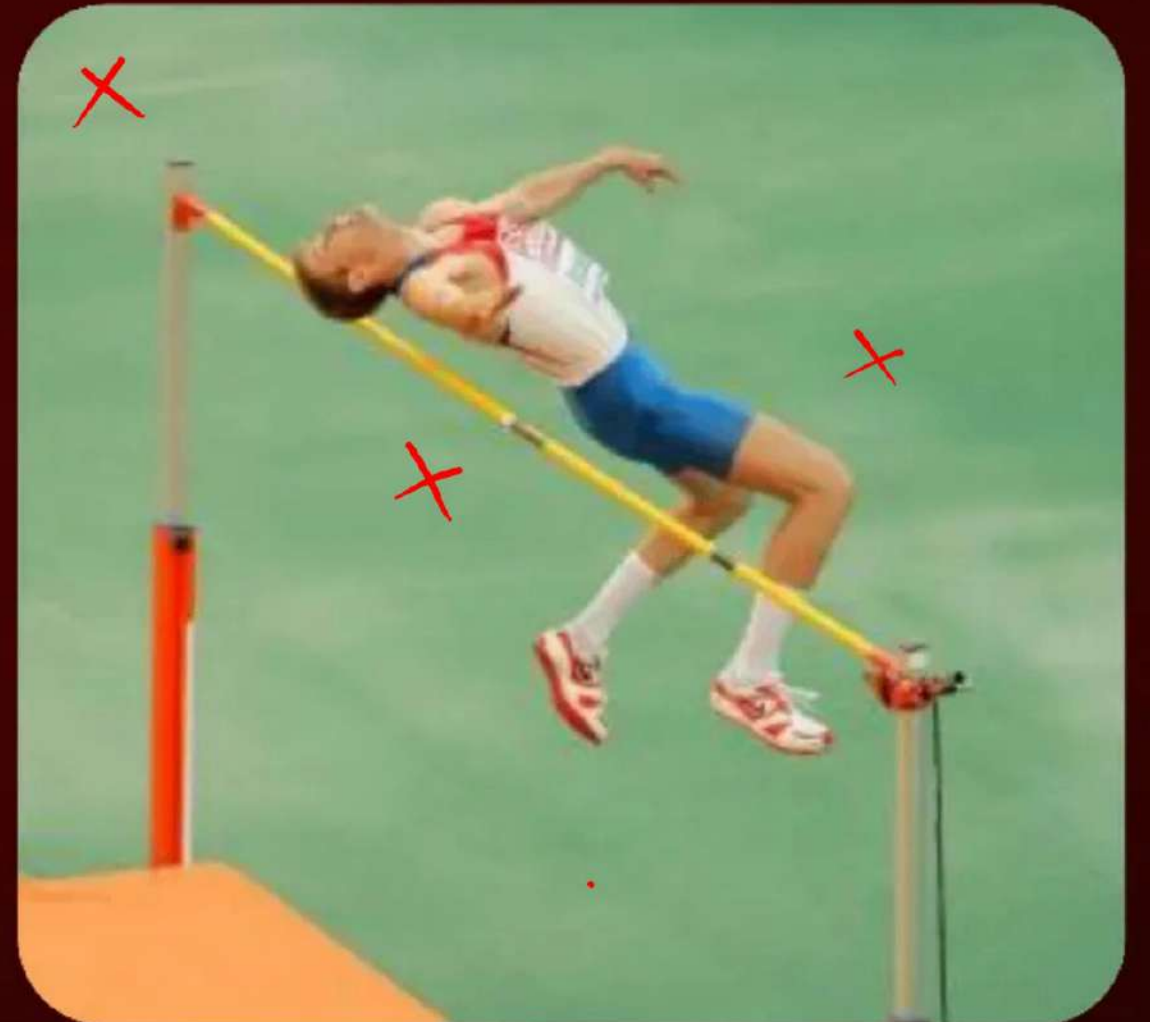


Application of Impulse

3. Karate Player while Breaking the tiles



$$\uparrow \uparrow F = \frac{dp}{dt} \downarrow$$



Q what is Impulse momentum theorem ? — 3 mark

2022

Ans

$$\uparrow F = \frac{dp}{dt} \downarrow$$

$$\underline{F dt} = dp$$

$$\downarrow \uparrow F = \frac{dp}{dt} \downarrow \uparrow$$

~~Impulse~~

$$\boxed{\underline{F dt} = dp}$$

→ Impulse momentum theorem.

$$\Delta p$$

$$\Delta p$$

The impulse ($F dt$) is change in momentum. This is velas.
Impulse momentum Theorem.

H/w

Q why newton 2nd law is klan Real law of motion? (2022, 23, 24)

SUMMARY

- ① Inertia
- ② momentum (p)
- ③ 2nd law
- ④ Impulse.
- ⑤ Application of Impulse
- ⑥ Impulse - momentum theorem.

Thank
You

$$F = \overset{(m)}{m} \frac{dv}{dt} + v \frac{dm}{dt} \overset{(0)}{0}$$

$$F = \overset{(m)}{m} \frac{dv}{dt} + v \times 0$$

$$F = \overset{(m)}{m} \frac{dv}{dt} \overset{(a)}{a} + 0$$

$$F = ma$$